Lemma 3.4. *Let $P = P(A, \mathbf{1})$, and let $\boldsymbol{c} \in (\mathbb{R}^d)^* \backslash \mathbb{0}$. If $\lambda > 0$ is small enough, then the linear function $\boldsymbol{c}^{(\lambda)}\boldsymbol{x}$ is in general position with respect to P, for*

$$\boldsymbol{c}^{(\lambda)} \;:=\; \boldsymbol{c} + (\lambda, \lambda^2, \ldots, \lambda^d).$$ □

3.2 Directing the Edges ("Linear Programming for Geometers")

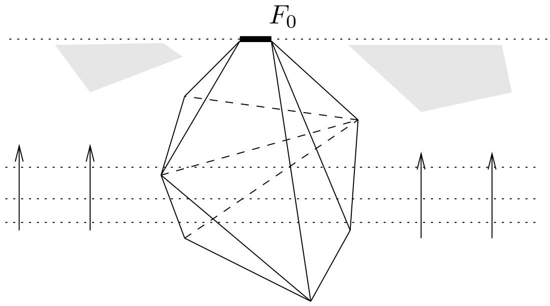
Definition 3.5. Let P be a convex polytope. The vertices and the edges of P form an abstract, finite, undirected, simple graph, called the *graph of* P and denoted by $G(P)$.

For every face $F \in L(P)$, we denote by $G(F)$ the *induced subgraph* of $G(P)$ on the subset $\mathrm{vert}(F) \subseteq \mathrm{vert}(P)$ of the vertices of $G(P)$, that is, the graph of all vertices in F, and all edges of P between them. This coincides with the graph of F, if F is itself considered as a polytope.

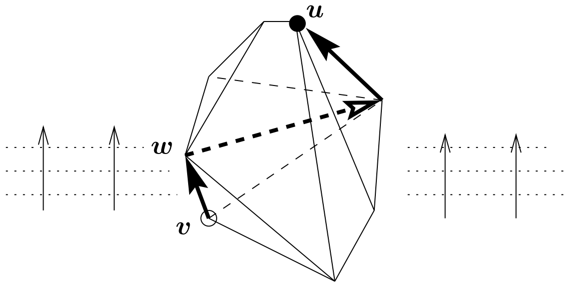
(In this whole course we need very little graph theory, only some terminology. When in doubt, look it up in any graph theory book. For that purpose, even a mediocre book would do. As for good ones, we recommend Bondy & Murty [123], Bollobás [121], or Tutte [551].)

We will consider *orientations* of $G(P)$, which assign a *direction* to every edge. An orientation is *acyclic* if there is no directed cycle in it. This implies (because all our graphs are finite) that there is a *sink*: a vertex that does not have an edge directed away from it. (Proof: Start at any vertex, and keep on walking along directed edges until you close a directed cycle or get stuck in a sink.)

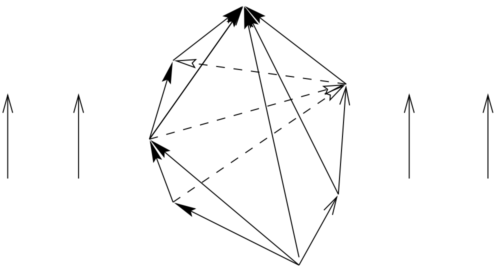
Linear programming is (in a geometer's version) the task to find a point $\boldsymbol{x}_0 \in P$ that maximizes a linear function $\boldsymbol{cx}$, that is, such that $\boldsymbol{cx}_0 = \max\{\boldsymbol{cx} : \boldsymbol{x} \in P\} =: c_0$. Now we easily see that the maximum is achieved in a vertex. In fact, $F_0 := \{\boldsymbol{x} \in P : \boldsymbol{cx} = c_0\}$ is a face of P, and thus every vertex of F_0 maximizes $\boldsymbol{cx}$.



Dantzig's *simplex algorithm* [174] in its "first phase" finds a vertex $\boldsymbol{v}$ of P. Then it proceeds to find a better vertex $\boldsymbol{w}$ that is a *neighbor* of $\boldsymbol{v}$. We use $N(\boldsymbol{v})$ to denote the set of neighbors of $\boldsymbol{v}$, that is, the set of all $\boldsymbol{w} \in \mathrm{vert}(P)$ such that $\mathrm{conv}\{\boldsymbol{v}, \boldsymbol{w}\}$ is an edge of P. This improvement step is iterated until the algorithm stops at an optimal vertex.



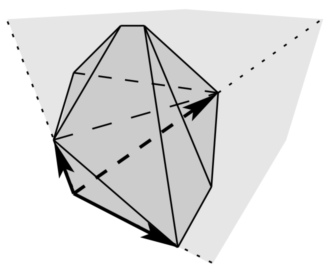
Now if $\boldsymbol{c}$ is in general position, then this gives us a well-defined way to direct the graph of P, by directing an edge $\mathrm{conv}\{\boldsymbol{v}_i, \boldsymbol{v}_j\}$ from $\boldsymbol{v}_i$ to $\boldsymbol{v}_j$ if $\boldsymbol{c}\boldsymbol{v}_i < \boldsymbol{c}\boldsymbol{v}_j$. (Because of the general position assumption, ties cannot occur.) We call this the *orientation of* $G(P)$ *induced by* $\boldsymbol{c}$.



With this construction *monotone paths* on P (edge paths for which the objective function increases strictly in each step) translate into directed paths in the orientation of $G(P)$ induced by $\boldsymbol{c}$.

Lemma 3.6. *Let $\boldsymbol{v} \in \mathrm{vert}(P)$ be a vertex, and let $N(\boldsymbol{v})$ be the set of its neighbors in $G(P)$. Then the cone (based at $\boldsymbol{v}$) spanned by the neighbors of $\boldsymbol{v}$ contains P:*

$$P \subseteq \boldsymbol{v} + \mathrm{cone}\{\boldsymbol{u} - \boldsymbol{v} : \boldsymbol{u} \in N(\boldsymbol{v})\}.$$



Proof. This follows from our proof of Proposition 2.4: the neighbors of $\boldsymbol{v}$ are in one-to-one correspondence with the vertices of the vertex figure $P/\boldsymbol{v}$, and thus it is equivalent to say that those vertices span a cone that contains P. But we have also seen there that every ray emanating from $\boldsymbol{v}$ to any other point $\boldsymbol{x} \in P$ contains a point of the vertex figure. This yields

$$\begin{aligned} P &\subseteq \{\boldsymbol{v} + t(\boldsymbol{u} - \boldsymbol{v}) : \boldsymbol{u} \in P/\boldsymbol{v},\ t \geq 0\} \\ &\subseteq \boldsymbol{v} + \mathrm{cone}\{\boldsymbol{u} - \boldsymbol{v} : \boldsymbol{u} \in \mathrm{vert}(P/\boldsymbol{v})\} \\ &= \boldsymbol{v} + \mathrm{cone}\{\boldsymbol{u} - \boldsymbol{v} : \boldsymbol{u} \in N(\boldsymbol{v})\}. \end{aligned}$$

□

Theorem 3.7. *If $\boldsymbol{cx}$ is a linear function in general position for P, then the orientation of $G(P)$ induced by $\boldsymbol{c}$ is acyclic, with a unique sink. This sink is the unique point in P where $\boldsymbol{cx}$ achieves its maximum.*

Proof. Along any directed path $\boldsymbol{v}_0, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ in $G(P)$, the value of $\boldsymbol{cx}$ increases strictly. Thus a directed path cannot return to its starting vertex, and there are no directed cycles. Therefore the induced orientation of $G(P)$ is acyclic, and it has a sink.

Now assume that $\boldsymbol{v}$ is a sink: then all of its neighbors $\boldsymbol{w} \in N(\boldsymbol{v})$ satisfy $\boldsymbol{cw} < \boldsymbol{cv}$. By Lemma 3.6 this implies that $\boldsymbol{cx} < \boldsymbol{cv}$ holds for all $\boldsymbol{x} \in P$ with $\boldsymbol{x} \neq \boldsymbol{v}$; that is, $\boldsymbol{v}$ maximizes $\boldsymbol{cx}$ over P, and it is the only point in P that achieves the maximum. □

This proves that for any starting vertex $\boldsymbol{v} \in \mathrm{vert}(P)$, and for any linear function $\boldsymbol{cx}$ that is in general position with respect to P, every strictly increasing edge path will eventually lead to the unique vertex that maximizes $\boldsymbol{cx}$ over P.

With this crude description, the problem of linear programming is, of course, not solved. This starts with the fact that, for efficient treatment, we have to consider bases and pivots instead of vertices and edges. (See Exercise 3.10 for a brief sketch.) Here we run into problems of degeneracy if the polytope is not simple, or if the linear function is not in general position. One way to treat this is through "perturbation," implicitly or explicitly. For example, if we know an interior point (this is not a natural assumption for practical problems!), then we can rewrite P as $P(A, \mathbf{1})$, and then (implicitly rather than explicitly) optimize over $P(A, \mathbf{1}^\lambda)$ for small enough $\lambda > 0$, which is nondegenerate. This leads to lexicographic pivot rules; see Chvátal [158, pp. 34–36]. Furthermore, to construct a simplex algorithm we have to determine "which edge to take"; this leads to the question of pivot rules. All this is combinatorial geometry. Later in the game, numerical questions dominate the picture.

Anyway, this discussion was only meant as a sketch of the geometric situation — a very simple and special picture of the world according to a discrete geometer.

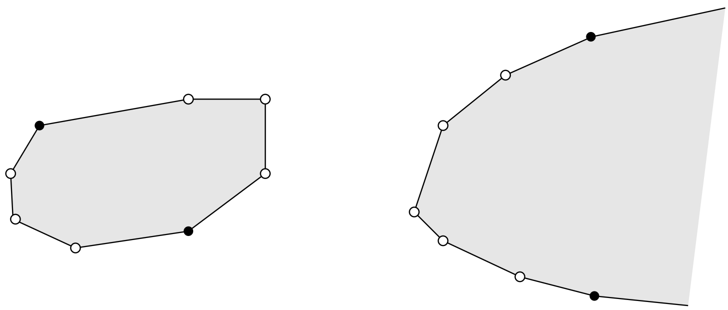
3.3 The Hirsch Conjecture

The *diameter* of a graph G will be denoted by $\delta(G)$: the smallest number δ such that any two vertices in G can be connected by a path with at most δ edges.

For $n > d \geq 2$, let $\Delta(d,n)$ be the maximal diameter of the graph of an d-dimensional polytope P with at most n facets. Similarly, let $\Delta_u(d,n)$ denote this maximal diameter in the unbounded case, for a d-dimensional pointed polyhedron P with at most n facets ($n \geq d \geq 2$). For example,

$$\Delta(2,n) \;=\; \lfloor \frac{n}{2} \rfloor, \qquad\qquad \Delta_u(2,n) \;=\; n-2.$$

Our sketch illustrates the extreme cases for $d = 2$ and $n = 8$.



It is a long-standing problem to determine the behavior of the function $\Delta(d,n)$. The value of $\Delta(d,n)$ is a lower bound for the number of iterations needed for the simplex algorithm with *any* pivot rule. Thus the question of whether $\Delta(d,n)$ grows polynomially in n and d is closely related to the question of whether there is any pivot rule for which the simplex algorithm is a strongly polynomial algorithm for linear programming; see [327, Sect. 3].

A notorious, very specific, question connected with the graphs of polytopes was first posed by Warren M. Hirsch in 1957 (see Dantzig [174, pp. 160, 168]) and has become known as the *Hirsch conjecture.*

Conjecture 3.8 (Hirsch conjecture). [174, p. 168]
For $n > d \geq 2$, let $\Delta(d,n)$ denote the largest possible diameter of the graph of a d-polytope with n facets. Then

$$\Delta(d,n) \leq n - d.$$

Is this plausible? Here are a few observations, most of them due to Klee & Walkup [331].